Real Numbers

1.2 The Fundamental Theorem of Arithmetic

In your earlier classes, you have seen that any natural number can be written as a product of its prime factors. For instance, $2 = 2$, $4 = 2 \times 2$, $253 = 11 \times 23$, and so on. Now, let us try and look at natural numbers from the other direction. That is, can any natural number be obtained by multiplying prime numbers? Let us see.

Take any collection of prime numbers, say 2, 3, 7, 11 and 23. If we multiply some or all of these numbers, allowing them to repeat as many times as we wish, we can produce a large collection of positive integers (In fact, infinitely many). Let us list a few :

$7 \times 11 \times 23 = 1771$ $\quad\quad$ $3 \times 7 \times 11 \times 23 = 5313$

$2 \times 3 \times 7 \times 11 \times 23 = 10626$ $\quad\quad$ $2^3 \times 3 \times 7^3 = 8232$

$2^2 \times 3 \times 7 \times 11 \times 23 = 21252$

and so on.

Theorem 1.1 (Fundamental Theorem of Arithmetic) : *Every composite number can be expressed (factorised) as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.*

2	48
2	24
2	12
2	6
3	3
	1

$48 = 2 \times 2 \times 2 \times 2 \times 3$

3	48
2	16
2	8
2	4
2	2
	1

$48 = 3 \times 2 \times 2 \times 2 \times 2$

$48 = 2 \times 2 \times 3 \times 2 \times 2$

$6, 20$

Q1 Find the LCM & HCF of 6 and 20.

2	6
3	3
	1

2	20
2	10
5	5
	1

$6 = 2 \times 3 = 2 \times 3$

$20 = 2 \times 2 \times 5 = 2^2 \times 5$

HCF = 2

LCM = $2^2 \times 3 \times 5$

= 60

HCF(6,20) × LCM(6,20) = 6×20

2 × 60

= 120

120

Example 2 : Find the LCM and HCF of 6 and 20 by the prime factorisation method.

Solution : We have : $6 = 2^1 \times 3^1$ and $20 = 2 \times 2 \times 5 = 2^2 \times 5^1$.

You can find HCF(6, 20) = 2 and LCM(6, 20) = $2 \times 2 \times 3 \times 5 = 60$, as done in your earlier classes.

Note that HCF(6, 20) = 2^1 = **Product of the smallest power of each common prime factor in the numbers.**

LCM (6, 20) = $2^2 \times 3^1 \times 5^1$ = **Product of the greatest power of each prime factor, involved in the numbers.**

From the example above, you might have noticed that HCF(6, 20) × LCM(6, 20) = 6 × 20. In fact, we can verify that **for any two positive integers *a* and *b*, HCF (*a*, *b*) × LCM (*a*, *b*) = *a* × *b*.** We can use this result to find the LCM of two positive integers, if we have already found the HCF of the two positive integers.

From the example above, you might have noticed that HCF(6, 20) × LCM(6, 20) = 6 × 20. In fact, we can verify that **for any two positive integers *a* and *b*, HCF (*a*, *b*) × LCM (*a*, *b*) = *a* × *b*.** We can use this result to find the LCM of two positive integers, if we have already found the HCF of the two positive integers.

For any two number a and b

HCF (a and b) =

LCM (a, b) =

Find the HCF and LCM of 336 and 54

2	336
2	168
2	84
2	42
3	21
7	7
	1

2	54
3	27
3	9
3	3
	1

$54 = 2^1 \times 3^3$

$336 = 2^4 \times 3 \times 7$

$HCF = 2^1 \times 3^1 = (6)$

Product of smallest power of each common factor

$HCF(54, 336) = 6$

$LCM(54, 336) = 3024$

$LCM = 2^4 \times 3^3 \times 7 = (3024)$

Product of greatest power of each prime factor

Example 3: Find the HCF of 96 and 404 by the prime factorisation method. Hence, find their LCM.

Sol^{n}

2	96
2	48
2	24
2	12
2	6
3	3
	1

2	404
2	202
101	101
	1

$96 = 2^5 \times 3$

$404 = 2^2 \times 101$

$HCF = 2^2 = 4$

$LCM = 2^5 \times 3 \times 101$

$= 9696$

OR

$LCM(96,404) \times HCF(96,404) = 96 \times 404$

$LCM(96,404) \times 4 = 96 \times 404$

$LCM(96,404) = \frac{96 \times 404}{4}$

$= 96 \times 101$

$= 9696$

Example 4 : Find the HCF and LCM of 6, 72 and 120, using the prime factorisation method.

$6 = 2 \times 3$

$72 = 2^3 \times 3^2$

$120 = 2^3 \times 3 \times 5$

$HCF = 2^1 \times 3^1$

$= 6$

$LCM = 2^3 \times 3^2 \times 5$

$= 8 \times 9 \times 5$

$= \boxed{360}$

$HCF \times LCM \overset{?}{=} 6 \times 72 \times 120$

$6 \times 360 \overset{?}{=} 51840$

$= 2160 \neq 51840$

EXERCISE 1.1

1. Express each number as a product of its prime factors:

(i) 140 (ii) 156 (iii) 3825 (iv) 5005 (v) 7429

2. Find the LCM and HCF of the following pairs of integers and verify that LCM × HCF = product of the two numbers.

   (i) 26 and 91 (ii) 510 and 92 (iii) 336 and 54

3. Find the LCM and HCF of the following integers by applying the prime factorisation method.

   (i) 12, 15 and 21 (ii) 17, 23 and 29 (iii) 8, 9 and 25

4. Given that HCF (306, 657) = 9, find LCM (306, 657).

$2^3 \times 3^2$ and $3^3 \times 2^2 \times 7$

HCF $2^2 \times 3^2$

5. Check whether 6^n can end with the digit 0 for any natural number n.

2 and 5

$6^n = (2 \times 3)^n$

$= 2^n \times 3^n$

$6^2 = 36$

$6^3 =$

$6^4 =$

$6^5 =$

$3^n \; 4^n \; 5^n$

$4^n = (2 \times 2)^n$

$= 2^n \times 2^n$

6. Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers.

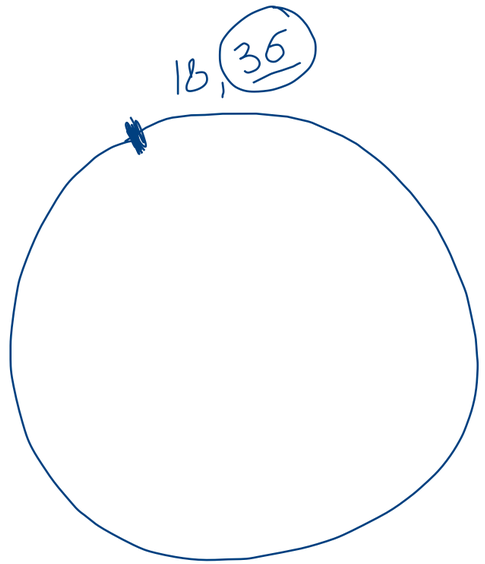
$7 \times 11 \times 13 + 13$

$13(7 \times 11 + 1)$

13×78

$5 \times (\quad)$

7. There is a circular path around a sports field. Sonia takes 18 minutes to drive one round of the field, while Ravi takes 12 minutes for the same. Suppose they both start at the same point and at the same time, and go in the same direction. After how many minutes will they meet again at the starting point?



Theorem 2

Let p is a prime number. If p divides a^2 then p divides a also.

$2 \qquad 4^2$

fractions V/S Rational

06/03/26

Qn Prove that $\sqrt{2}$ is irrational.

Sol^{n} Let's assume that $\sqrt{2}$ is a rational number, which can be written in the form $\frac{p}{q}$ where $q \neq 0$ and p, q are integers. Also p and q are co-prime numbers.

$$\sqrt{2} = \frac{p}{q} \Rightarrow \sqrt{2}\,q = p$$

Squaring both sides, we get

$$(\sqrt{2}q)^2 = p^2$$

$$2q^2 = p^2 \quad \text{— (i)}$$

$q^2 = \frac{p^2}{2} \Rightarrow p^2$ is divisible by 2. — (ii)

According to fundamental theorem of arithmetic

If p is a prime number where a^2 is divisible by p then a is also divisible by p

from (ii) we can say p is also divisible by 2. — (iii)

$\frac{p}{2} = r$

$\Rightarrow p = 2r$

Squaring both sides

$p^2 = 4r^2$

Substitute $p^2 = 4r^2$ in (i)

$2q^2 = 4r^2$

$q^2 = 2r^2$

$\frac{q^2}{2} = r^2 \Rightarrow q^2$ is divisible by 2

q is also divisible by 2. (iv)

from (iii) and (iv)

we observe that p and q both have a common factor 2.

which contradicts the fact that p and q are co-prime

It is occuring due to our wrong assumption that $\sqrt{2}$ is a rational number

So, $\sqrt{2}$ is irrational number.

$$\sqrt{5} = \frac{p}{q}$$

$$\Rightarrow \sqrt{5}q = p$$

$$\Rightarrow 5q^2 = p^2$$

$$q^2 = \frac{p^2}{5}$$

(p^2 is divisible by 5)

p is also divisible by 5

$$\frac{p}{5} = r$$

$$p = 5r$$

$$p^2 = 25r^2$$

$$5q^2 = 25r^2$$

$$q^2 = 5r^2 \Rightarrow \frac{q^2}{5} = r^2$$

q^2 is divisible by 5

q is also divisible by 5.

Q2 Prove that $3+2\sqrt{5}$ is irrational.

$$3+2\sqrt{5}=\frac{p}{q}$$

$$2\sqrt{5}=\frac{p}{q}-3$$

$$\sqrt{5}=\frac{\frac{p}{q}-3}{2} \quad \text{or} \quad \frac{\frac{p}{q}-3}{2}$$

$\frac{1}{2}\left(\frac{p}{q}-3\right)$ will be a rational number for every integral value of p and q.

$\therefore \sqrt{5}$ is also a rational number.

But it contradicts the fact that $\sqrt{5}$ is an irrational number.

$$\frac{1}{\sqrt{2}}=\frac{p}{q}$$

$$\sqrt{2}p=q$$

$$\sqrt{2}=\frac{q}{p}$$

24. If H.C.F. of $135x^2$ and $189x^3$ is 108, then find the value of x.

3	135
3	45
3	15
5	5
	1

3	189
3	63
3	21
7	7
	1

$135x^2 = 3^3 \times 5 \times x^2$

$189x^3 = 3^3 \times 7 \times x^3$

$\boxed{HCF} = 3^3 \times x^2$

$27x^2 = 108$

$x^2 = \frac{108}{27} = 4$

$x^2 = 4$

$\boxed{x = 2}$

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Practice Questions

Q1. If HCF(306, 657) = 9, then LCM(306, 657) = ____.

Q2. If $2^3 \times 3^a \times b \times 7$ is the prime factorisation of 2520, then $5a+2b$ = ?

Q3. If the HCF and LCM of two integers 'a' and 'b' are x and y.

Then find the value of $\frac{x^2 y^2}{a^2 b^2}$

Q4. Find the sum of exponents of the prime factors in the prime factorisation of 196.

Q5. Find HCF(a, b) and LCM(a, b) if $a = pq^2$ and $b = p^3 q$

Q6. Prove that $\frac{2+\sqrt{3}}{5}$ is an irrational, given that $\sqrt{3}$ is an irrational number.

① HCF × LCM = 306 × 657

9 × LCM = 306 × 657

$LCM = \frac{306 \times 657}{9}$

$= 34 \times 657$

$= 22338$

② 2 | 2520

2 | 1260

2 | 630

3 | 315

3 | 105

5 | 35

7 | 7

1

$= 2^3 \times 3^2 \times 5 \times 7$

$2^3 \times 3^a \times b \times 7$

$a = 2$

$b = 5$

$5a + 2b = 5(2) + 2(5)$

$= 10 + 10$

$= 20$

③ HCF × LCM = product of two numbers

$x \times y = a \times b$

$xy = ab$

$x^2y^2 = a^2b^2$

$\frac{x^2y^2}{a^2b^2} = 1$

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Poachce Questions

Q1. If HCF(306, 657) = 9, then LCM(306, 657) = ____.

Q2. If $2^3 \times 3^a \times b \times 7$ is the prime factorisation of 2520, then $5a + 2b = ?$

Q3. If the HCF and LCM of two integers 'a' and 'b' are x and y

Then find the value of $\frac{x^2y^2}{a^2b^2} = \frac{a^2b^2}{a^2b^2} = 1$

Q4. Find the sum of exponents of the prime factors in the prime factorisation of 196

Q5. Find HCF(a, b) and LCM(a, b) if $a = pq^2$ and $b = p^3q$

Q6. Prove that $\frac{2+\sqrt{3}}{5}$ is an irrational, given that $\sqrt{3}$ is an irrational number.

⑤ $a = p^1q^2$

$b = p^3q$

HCF = pq

LCM = p^3q^2

$a = p^3q^2z$

$b = pq$

LCM = p^3q^2z

HCF = pq

④ 2 | 196

2 | 98

7 | 49

7 | 7

1

$196 = 2^2 \times 7^2$

Exponents of 2 and 7 are 2 and 2

So 2 + 2 = 4

⑥ $\frac{2+\sqrt{3}}{5} = \frac{p}{q}$

$2 + \sqrt{3} = \frac{5p}{q}$

$\sqrt{3} = \frac{5p}{q} - 2$

$\frac{a+b}{c} = \frac{d}{e}$

$\frac{a}{c} + \frac{b}{c} = \frac{d}{e}$

$\frac{b}{c} = \frac{d}{e} - \frac{a}{c}$

1.4 Summary

In this chapter, you have studied the following points:

1. The Fundamental Theorem of Arithmetic :

   Every composite number can be expressed (factorised) as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.

2. If p is a prime and p divides a^2, then p divides a, where a is a positive integer.

3. To prove that $\sqrt{2}, \sqrt{3}$ are irrationals.

(iii)

$4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3}$

$4x(\sqrt{3}x + 2) - \sqrt{3}(\sqrt{3}x + 2)$

$(\sqrt{3}x + 2)(4x - \sqrt{3}) = 0$

$\sqrt{3}x + 2 = 0$ or $4x - \sqrt{3} = 0$

$x = -\frac{2}{\sqrt{3}}$ (α) $\quad x = \frac{\sqrt{3}}{4}$ (β)

$\alpha + \beta = \frac{-2}{\sqrt{3}} + \frac{\sqrt{3}}{4} = \frac{-8+3}{4\sqrt{3}} = \frac{-5}{4\sqrt{3}}$ $\quad \frac{-b}{a} = \frac{-5}{4\sqrt{3}}$

$\alpha\beta = \frac{-2}{\sqrt{3}} \times \frac{\sqrt{3}}{4} = \frac{-1}{2}$ $\quad \frac{c}{a} = \frac{-2\sqrt{3}}{4\sqrt{3}} = \frac{-1}{2}$

$O \times O = -24$

$O + O = 5$

$8, -3$

(2) i) $\alpha + \beta = -\frac{b}{a}$ $\quad \alpha\beta = \frac{c}{a}$

$\alpha^2 + \beta^2 = ?$

$(\alpha+\beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$

$\left(-\frac{b}{a}\right)^2 = \alpha^2 + \beta^2 + 2\left(\frac{c}{a}\right)$

$\frac{b^2 - 2ac}{a^2} = \frac{b^2}{a^2} - \frac{2c}{a} = \alpha^2 + \beta^2$

PRACTICE TEST

Q1. Find the zeroes of the polynomial and verify the relation of zeroes & coefficient.

(i) $6x^2 - x - 2$

(ii) $6x^2 - 3$

(iii) $4\sqrt{3}x^2 + 5x - 2\sqrt{3}$

Q2. If α & β are the zeroes of $f(x) = ax^2 + bx + c$, then evaluate

(i) $\alpha^2 + \beta^2$

(ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

(ii) $\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{\frac{b^2 - 2ac}{a^2}}{\frac{c}{a}} = \frac{b^2 - 2ac}{ac}$

$\frac{b^2 - 2ac}{a^2} \times \frac{a}{c}$

$6x^2 - 3 = 0$

$6x^2 = 3$

$x^2 = \frac{3}{6} = \frac{1}{2}$

$x^2 = \frac{1}{2}$

$x = \frac{1}{\sqrt{2}}$ or $-\frac{1}{\sqrt{2}}$

$x = \pm\frac{1}{\sqrt{2}}$

$\alpha = \frac{1}{\sqrt{2}}$, $\beta = \frac{-1}{\sqrt{2}}$

$\alpha + \beta = \frac{1}{\sqrt{2}} + \left(\frac{-1}{\sqrt{2}}\right) = 0$

$\alpha\beta = \frac{1}{\sqrt{2}} \times \left(\frac{-1}{\sqrt{2}}\right) = \frac{-1}{2}$

$\frac{-b}{a} = \frac{-(0)}{6} = 0$

$\frac{c}{a} = \frac{-3}{6} = \frac{-1}{2}$

$6x^2 + 0x - 3 = 0$

$\sqrt{18}, \sqrt{18}$

$O \times O = -18$

$O + O = 0$

$\sqrt{18} = \sqrt{2 \times 3 \times 3} = 3\sqrt{2}$

Q1 If $a = 2^4 \times 3^2 \times 5$

$b = 2^2 \times 3 \times 5$

$c = 3^n \times 5^2$

and $LCM(a,b,c) = 2^4 \times 3^4 \times 5^2$

find n

Q2. If $a = pq^2$, $b = p^3q$

then $LCM(a,b) = p^3q^2$

$HCF(a,b) =$

Q3. Prove that $\sqrt{7}$ is irrational.

Lets assume, to the contrary, that $\sqrt{7}$ is rational where it can be written in $\frac{p}{q}$ form, where $q \neq 0$ and p&q are integers also, p&q are coprime numbers, and in the standard form of rational numbers.

$$\frac{\sqrt{7}}{1} = \frac{p}{q}$$

$$\sqrt{7}q = p$$

Squaring both sides

$$(\sqrt{7}q)^2 = p^2$$

$$7q^2 = p^2$$

So using fundamental theorem

Prove that $\sqrt{7}$ is irrational.

$$\sqrt{7} = \frac{p}{q}$$

$$\sqrt{7}q = p$$

$$7q^2 = p^2$$

$q^2 = \frac{p^2}{7}$ → p^2 is divisible by 7

p is also divisible by 7. {Theorem}

$$\frac{p}{7} = r$$

$$p = 7r$$

$$p^2 = 49r^2$$

$$7q^2 = 49r^2$$

$$q^2 = 7r^2$$

$\frac{q^2}{7} = r^2$ → q^2 is divisible

so q is also divisible by 7.

$$3 + 2\sqrt{5} = \frac{p}{q}$$

$$2\sqrt{5} = \frac{p}{q} - 3$$

$$2\sqrt{5} = \frac{p - 3q}{q}$$

$$\sqrt{5} = \frac{p - 3q}{2q}$$

Qn. Prove that $5-\sqrt{3}$ is irrational.

$5-\sqrt{3} = \frac{p}{q}$

$5 - \frac{p}{q} = \sqrt{3}$

$\frac{5q-p}{q} = \sqrt{3}$

$5-\sqrt{3} = \frac{p}{q}$

$-\sqrt{3} = \frac{p}{q} - 5$

$-\sqrt{3} = \frac{p-5q}{q}$

$\sqrt{3} = -\left(\frac{p-5q}{q}\right)$

$\sqrt{3} = \frac{5q-p}{q}$

144, 180, 192

2	192
2	96
2	48
2	24
2	12
2	6
3	2
	1

$192 = 2^2 \times 3 \times 2^3 \times 3$

$144 \Rightarrow 2^4 \times 3^2$

$180 \Rightarrow 2^2 \times 3^2 \times 5$

$92 \Rightarrow 2^6 \times 3$

2	180
2	90
5	45
3	9
3	3
	1

HCF $= 2^2 \times 3$

LCM $= 2^6 \times 3^2 \times 5^1$

$= 2880$

LCM $\Rightarrow 2^6 \times 3^2 \times 5$

HCF $\Rightarrow 2^2 \times 3$

$\Rightarrow 12$

$2880 \leftarrow$